Vyomesh Kumar
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SUMMARY

Computer Science student at USF (GPA: 3.83, Presidential Scholar) pursuing software engineering and AI/ML. Built and deployed Python applications using LLM APIs, REST APIs, OAuth, cloud scheduling, automated testing, and persistent data. Currently developing a predictive maintenance machine learning model using NASA turbofan engine degradation data to predict Remaining Useful Life (RUL) from multivariate sensor data. Skilled in Python application development, API integration, debugging, and collaborative problem-solving; also developing a voice-controlled AI robot with USF ACM. Seeking Summer 2027 software engineering and AI/ML internships.

EDUCATION

University of South Florida | Tampa, FL

B.S. Computer Science | GPA: 3.83/4.0 | Green & Gold Presidential Scholarship | Expected Spring 2028

Relevant Coursework: Computer Organization, Program Design, Programming Concepts, Discrete Structures, Probability & Statistics with Calculus, Linear Systems

PROJECTS

NASA Turbofan Engine Remaining Useful Life (RUL) Prediction | Python, scikit-learn, Pandas, Machine Learning - In Progress

• Developing an end-to-end predictive maintenance machine learning pipeline using NASA's turbofan engine degradation dataset to predict Remaining Useful Life (RUL) from multivariate time-series sensor data and operational cycles.

• Implementing data preprocessing, feature engineering, and Random Forest regression using Python, Pandas, NumPy, and scikit-learn; evaluating model generalization through GroupKFold cross-validation and Mean Absolute Error (MAE).

• Planning to optimize model performance through hyperparameter tuning and deploy an interactive Streamlit web application that enables users to input engine sensor data and receive RUL predictions for predictive maintenance decisions.

LinkedIn Career Agent | Python, Claude API, Apify, Streamlit, Pytest, JSON github.com/vyomeshk2006-code/linkedin-Agent

• Built and deployed an AI-enabled web application with 10 user-facing features, including profile audits, competitor comparisons, internship eligibility checks, and career roadmap generation; integrated LLM and data APIs with persistent JSON session memory and application logic to maintain conversational context across user interactions.

• Developed 23 pytest test cases using mocked API calls to test input validation, session memory, conversation history, and agent features; identified and fixed a production input-validation bug through testing.

Email Summarizer | Python, Claude API, Gmail API, OAuth2.0, APScheduler github.com/vyomeshk2006-code/Email-Summarizer

• Built and deployed production code for an AI-powered email service integrating Gmail and Claude APIs to automate prioritized daily reports summarizing messages from the previous 24 hours, emphasizing reliable automation and user-focused output.

• Designed an automated backend workflow using OAuth authentication and APScheduler; deployed to Railway for reliable 24/7 cloud execution without local intervention, while debugging API integrations and scheduling behavior across external services.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL
Machine Learning & Data Science: scikit-learn, Pandas, NumPy, PyTorch, TensorFlow, Random Forest, Regression, Feature Engineering, Predictive Modeling, Time-Series Analysis, Cross-Validation, Model Evaluation
Tools & Platforms: Git, GitHub, Streamlit, Railway, pytest, Google Apps Script, Apify, OAuth 2.0, Notion
Software Engineering: API Integration, Workflow Automation, Full-Stack Development, Software Testing, Debugging, Version Control, Problem-Solving, REST APIs

CERTIFICATIONS

• Machine Learning Specialization (Courses 1 & 2) – Supervised Machine Learning: Regression and Classification; Advanced Learning Algorithms | DeepLearning.AI / Stanford Online (Coursera)
• Advanced Python - LinkedIn Learning

LEADERSHIP & ACTIVITIES

Software Lead - ACM Project Hub Team, USF Chapter Aug 2026 - Present

• Developing software for a voice-controlled AI robot using Python, Raspberry Pi, and OpenAI technologies, focusing on voice interaction, AI integration, and movement control; collaborating with the Project Manager and programmers on implementation. Contributing to feature integration, testing, debugging, and team coordination using GitHub and Notion.

Resident Assistant - University of South Florida Housing Aug 2026 – Present

• Support 40+ residents by resolving issues independently, communicating effectively, and adapting to changing priorities.